

JOIN Exercise

BRIGHTLEARN

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Objective

The objective of this exercise is to develop an understanding of how INNER JOIN, LEFT JOIN, and FULL OUTER JOIN are used to combine data from multiple tables. Also to strengthen practical SQL skills by writing queries that retrieve, match, and identify missing relationships between tables.

Question 1

```
1  -- PART A - Inner JOIN
2  -- Question 1 Users who have subscriptions
3  SELECT
4      u.user_id,
5      u.user_name,
6      s.subscription_id,
7      s.start_date
8  FROM workspace.default.users u
9  INNER JOIN workspace.default.subscriptions s
10     ON u.user_id = s.user_id;
```

Add parameter

	user_id	user_name	subscription_id	start_date
1	1	Nomvula	503	2026-03-10T00:00:00.000+00:...
2	2	David	502	2026-02-01T00:00:00.000+00:...
3	3	Anele	505	2026-04-05T00:00:00.000+00:...
4	1	Nomvula	501	2026-01-15T00:00:00.000+00:...

Question 2

```
12  -- Question 2 Subscriptions with their plans
13  SELECT
14      s.subscription_id,
15      s.user_id,
16      p.plan_name,
17      p.monthly_price
18  FROM workspace.default.subscriptions s
19  INNER JOIN workspace.default.plans p
20     ON s.plan_id = p.plan_id;
```

Add parameter

	subscription_id	user_id	plan_name	monthly_price
1	501	1	Basic	79
2	504	6	Standard	129
3	503	1	Premium	199
4	505	3	Family	249
5	502	2	Standard	129

Question 3

```
22 | --Question 3 Viewing sessions with matching shows
23 | SELECT
24 |     v.session_id,
25 |     v.user_id,
26 |     sh.show_title,
27 |     sh.genre,
28 |     v.watch_minutes
29 | FROM workspace.default.viewing_sessions v
30 | INNER JOIN workspace.default.shows sh
31 |     ON v.show_id = sh.show_id;
32 |
```

Add parameter

	1 ² ₃ session_id	1 ² ₃ user_id	1 ⁰ ₀ show_title	1 ⁰ ₀ genre	1 ² ₃ watch_minutes
1	904	7	Comedy Hour	Comedy	20
2	903	1	Crime Time	Drama	60
3	902	2	Tech Tales	Documentary	30
4	905	3	Wild Earth	Documentary	90
5	901	1	Comedy Hour	Comedy	45

Question 4

```
33 | -- Question 4 Viewing sessions with matching users
34 | SELECT
35 |     u.user_name,
36 |     u.country,
37 |     v.session_id,
38 |     v.show_id,
39 |     v.watch_minutes
40 | FROM workspace.default.users u
41 | INNER JOIN workspace.default.viewing_sessions v
42 |     ON u.user_id = v.user_id;
```

Add parameter

	1 ⁰ ₀ user_name	1 ⁰ ₀ country	1 ² ₃ session_id	1 ² ₃ show_id	1 ² ₃ watch_minutes
1	Nomvula	Johannesburg	903	702	60
2	David	Cape Town	902	703	30
3	Anele	Durban	905	705	90
4	Nomvula	Johannesburg	901	701	45

Question 5

```
44 | --Question 5 Users, subscriptions and plans
45 | SELECT
46 |     u.user_name,
47 |     u.country,
48 |     p.plan_name,
49 |     p.monthly_price,
50 |     s.start_date
51 | FROM users u
52 | INNER JOIN workspace.default.subscriptions s
53 |     ON u.user_id = s.user_id
54 | INNER JOIN workspace.default.plans p
55 |     ON s.plan_id = p.plan_id;
```

Add parameter

	1 ⁰ ₀ user_name	1 ⁰ ₀ country	1 ⁰ ₀ plan_name	1 ² ₃ monthly_price	1 ⁰ ₀ start_date
1	Nomvula	Johannesburg	Premium	199	2026-03-10T00:00:00.000+00:...
2	David	Cape Town	Standard	129	2026-02-01T00:00:00.000+00:...
3	Anele	Durban	Family	249	2026-04-05T00:00:00.000+00:...
4	Nomvula	Johannesburg	Basic	79	2026-01-15T00:00:00.000+00:...

Question 6

```
--PART B - LEFT JOIN
--Question 6 - All users and subscriptions
SELECT
  u.user_id,
  u.user_name,
  s.subscription_id,
  s.start_date
FROM workspace.default.users u
LEFT JOIN workspace.default.subscriptions s
  ON u.user_id = s.user_id;
```

Add parameter

	user_id	user_name	subscription_id	start_date
1	1	Nomvula	503	2026-03-10T00:00:00.000+00:...
2	2	David	502	2026-02-01T00:00:00.000+00:...
3	3	Anele	505	2026-04-05T00:00:00.000+00:...
4	4	Kabelo	null	null
5	5	Lerato	null	null
6	1	Nomvula	501	2026-01-15T00:00:00.000+00:...

Question 7

```
--Question 7 All plans and subscriptions
SELECT
  p.plan_id,
  p.plan_name,
  s.subscription_id,
  s.user_id
FROM workspace.default.plans p
LEFT JOIN workspace.default.subscriptions s
  ON p.plan_id = s.plan_id;
--Question 8
```

Add parameter

	plan_id	plan_name	subscription_id	user_id
1	10	Basic	501	1
2	11	Standard	504	6
3	12	Premium	503	1
4	13	Family	505	3
5	14	Mobile	null	null
6	11	Standard	502	2

Question 8

```
--Question 8 All shows and viewing sessions
SELECT
  sh.show_id,
  sh.show_title,
  v.session_id,
  v.watch_minutes
FROM workspace.default.shows sh
LEFT JOIN workspace.default.viewing_sessions v
  ON sh.show_id = v.show_id;
```

Add parameter

	show_id	show_title	session_id	watch_minutes
1	701	Comedy Hour	904	20
2	702	Crime Time	903	60
3	703	Tech Tales	902	30
4	704	Cooking Lab	null	null
5	705	Wild Earth	905	90
6	701	Comedy Hour	901	45

Question 9

```
88 | --Question 9 All viewing sessions with users
89 | SELECT
90 |     v.session_id,
91 |     v.show_id,
92 |     v.watch_minutes,
93 |     u.user_id,
94 |     u.user_name
95 | FROM workspace.default.viewing_sessions v
96 | LEFT JOIN workspace.default.users u
97 |     ON v.user_id = u.user_id;
98 |
```

Add parameter

	session_id	show_id	watch_minutes	user_id	user_name
1	901	701	45	1	Nomvula
2	902	703	30	2	David
3	903	702	60	1	Nomvula
4	904	701	20	null	null
5	905	705	90	3	Anele

Question 10

```
99 | --Question 10 All users and their plans
100 | SELECT
101 |     u.user_name,
102 |     u.country,
103 |     p.plan_name,
104 |     p.monthly_price
105 | FROM workspace.default.users u
106 | LEFT JOIN workspace.default.subscriptions s
107 |     ON u.user_id = s.user_id
108 | LEFT JOIN workspace.default.plans p
109 |     ON s.plan_id = p.plan_id;
110 |
```

Add parameter

	user_name	country	plan_name	monthly_price
1	Nomvula	Johannesburg	Premium	199
2	David	Cape Town	Standard	129
3	Anele	Durban	Family	249
4	Kabelo	Pretoria	null	null
5	Lerato	Port Elizabeth	null	null
6	Nomvula	Johannesburg	Basic	79

Question 11

```
111 | --PART C FULL OUTER JOIN
112 | -- Question 11 FULL OUTER JOIN users and subscriptions
113 | SELECT
114 |     COALESCE(u.user_id, s.user_id) AS user_id,
115 |     u.user_name,
116 |     s.subscription_id,
117 |     s.start_date
118 | FROM workspace.default.users u
119 | FULL OUTER JOIN workspace.default.subscriptions s
120 |     ON u.user_id = s.user_id;
121 |
```

Add parameter

	user_id	user_name	subscription_id	start_date
1	3	Anele	505	2026-04-05T00:00:00.000+00:00
2	4	Kabelo	null	null
3	2	David	502	2026-02-01T00:00:00.000+00:00
4	5	Lerato	null	null
5	1	Nomvula	503	2026-03-10T00:00:00.000+00:00
6	1	Nomvula	501	2026-01-15T00:00:00.000+00:00
7	6	null	504	2026-03-20T00:00:00.000+00:00

Question 12

```
122 -- Question 12 FULL OUTER JOIN plans and subscriptions
123 SELECT
124     COALESCE(p.plan_id, s.plan_id) AS plan_id,
125     p.plan_name,
126     s.subscription_id,
127     s.user_id
128 FROM workspace.default.plans p
129 FULL OUTER JOIN workspace.default.subscriptions s
130     ON p.plan_id = s.plan_id;
131
```

Add parameter

	plan_id	plan_name	subscription_id	user_id
1	11	Standard	504	6
2	14	Mobile	null	null
3	13	Family	505	3
4	12	Premium	503	1
5	10	Basic	501	1
6	11	Standard	502	2

Question 13

```
132 -- Question 13 FULL OUTER JOIN shows and viewing sessions
133 SELECT
134     COALESCE(sh.show_id, v.show_id) AS show_id,
135     sh.show_title,
136     v.session_id,
137     v.watch_minutes
138 FROM workspace.default.shows sh
139 FULL OUTER JOIN workspace.default.viewing_sessions v
140     ON sh.show_id = v.show_id;
141
142
```

Add parameter

	show_id	show_title	session_id	watch_minutes
1	704	Cooking Lab	null	null
2	705	Wild Earth	905	90
3	701	Comedy Hour	904	20
4	703	Tech Tales	902	30
5	702	Crime Time	903	60
6	701	Comedy Hour	901	45

Question 14

```
142 --Question 14 FULL OUTER JOIN users and sessions
143 SELECT
144     COALESCE(u.user_id, v.user_id) AS user_id,
145     u.user_name,
146     v.session_id,
147     v.show_id,
148     v.watch_minutes
149 FROM workspace.default.users u
150 FULL OUTER JOIN workspace.default.viewing_sessions v
151     ON u.user_id = v.user_id;
152
153
```

Add parameter

	user_id	user_name	session_id	show_id	watch_minutes
1	3	Anele	905	705	90
2	2	David	902	703	30
3	7	null	904	701	20
4	1	Nomvula	901	701	45
5	1	Nomvula	903	702	60
6	4	Kabelo	null	null	null
7	5	Lerato	null	null	null

Question 15

```
--  
153 -- Question 15 Users, subscriptions and plans using FULL OUTER JOIN  
154 SELECT  
155     COALESCE(u.user_id, s.user_id) AS user_id,  
156     u.user_name,  
157     s.subscription_id,  
158     COALESCE(p.plan_id, s.plan_id) AS plan_id,  
159     p.plan_name  
160 FROM workspace.default.users u  
161 FULL OUTER JOIN workspace.default.subscriptions s  
162     ON u.user_id = s.user_id  
163 FULL OUTER JOIN workspace.default.plans p  
164     ON s.plan_id = p.plan_id;  
165
```

Add parameter

Table									
1 ²	user_id	1 ⁰	user_name	1 ³	subscription_id	1 ³	plan_id	1 ⁰	plan_name
1		2	David		502	11	Standard		
2		6	null		504	11	Standard		
3		3	Anele		505	13	Family		
4		1	Nomvula		503	12	Premium		
5		4	Kabelo	null		null	null		
6		5	Lerato	null		null	null		
7		1	Nomvula	501		10	Basic		
8		null	null		null		14	Mobile	

Bonus Question 1

```
--  
166 -- BONUS QUESTION  
167 --BONUS Question 1: Which users have not subscribed to any plan?  
168 SELECT  
169     u.user_id,  
170     u.user_name,  
171     u.country  
172 FROM workspace.default.users u  
173 LEFT JOIN workspace.default.subscriptions s  
174     ON u.user_id = s.user_id  
175 WHERE s.subscription_id IS NULL;  
176
```

Add parameter

Table			
	1 ² user_id	1 ⁰ user_name	1 ⁰ country
1	4	Kabelo	Pretoria
2	5	Lerato	Port Elizabeth

Bonus Question 2

```
177 --BONUS Question 2: Which subscriptions reference users that do not exist?  
178 SELECT  
179     s.subscription_id,  
180     s.user_id,  
181     s.plan_id,  
182     s.start_date  
183 FROM workspace.default.subscriptions s  
184 LEFT JOIN workspace.default.users u  
185     ON s.user_id = u.user_id  
186 WHERE u.user_id IS NULL;  
187  
188
```

Add parameter

Table +				
	subscription_id	user_id	plan_id	start_date
1	504	6	11	2026-03-20T00:00:00.000+00:00...

Bonus Question 3

```
187
188 --BONUS Question 3: Which shows have never been watched?
189 SELECT
190     sh.show_id,
191     sh.show_title,
192     sh.genre
193 FROM workspace.default.shows sh
194 LEFT JOIN workspace.default.viewing_sessions v
195     ON sh.show_id = v.show_id
196 WHERE v.session_id IS NULL;
197
198
```

Add parameter

Table ▾ +			
	¹ ₃ show_id	^A ₀ show_title	^A ₀ genre
1	704	Cooking Lab	Lifestyle

Bonus Question 4

```
198 --BONUS Question 4: Which viewing sessions reference shows that do not exist?
199 SELECT
200     v.session_id,
201     v.user_id,
202     v.show_id,
203     v.watch_minutes
204 FROM workspace.default.viewing_sessions v
205 LEFT JOIN workspace.default.shows sh
206     ON v.show_id = sh.show_id
207 WHERE sh.show_id IS NULL;
208
209
```

Add parameter

Table ▾ +			
	¹ ₃ session_id	¹ ₃ user_id	¹ ₃ show_id
			¹ ₃ watch_minutes

No rows returned

Bonus Question 5

```
209 --BONUS Question 5: Which plans have no subscribers?
210 SELECT
211     p.plan_id,
212     p.plan_name,
213     p.monthly_price
214 FROM workspace.default.plans p
215 LEFT JOIN workspace.default.subscriptions s
216     ON p.plan_id = s.plan_id
217 WHERE s.subscription_id IS NULL;
218
```

Add parameter

Table ▾ +			
	¹ ₃ plan_id	^A ₀ plan_name	¹ ₃ monthly_price
1	14	Mobile	59